

Original study

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Seasonal and daily activity patterns of mammals in the colony of great cormorants

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Abstract: Despite extreme changes of ecosystems made by breeding colonies of great cormorants (*Phalacrocorax carbo*), these territories are still used by mammals. We present results of the analysis of mammal seasonal and daily activity patterns, registered by camera traps in two colonies of the great cormorants in Lithuania. Red foxes and raccoon dogs mainly visited colonies in the cormorant breeding time, April to July. In the inland colony of great cormorants in Lukštas Lake, average red fox relative shooting frequency in cormorant breeding time considerably exceeded that in non-breeding time (on average, 41.8 and 5.9 photos/100 days respectively). In the peninsular colony of great cormorants in Juodkrantė average relative shooting frequency of red fox in the breeding time was 7.8 versus 3.4 photos/100 days in non-breeding time, relative shooting frequencies of raccoon dog were 50.0 versus 1.3 photos/100 days, respectively. Daily activity patterns of wild boar and red fox in both colonies, as well as activity patterns of raccoon dog in Lukštas were related to the activity of cormorants with various degree of significance.

Keywords: activity differences; camera traps; cormorant colony; mammals.

1 Introduction

Great cormorants are able to induce drastic changes of the ecosystem in the territory of the breeding colonies (Kameda et al. 2006). A single bird consumes 400 g fish per day and deposits 20–50 g of faeces. Great cormorants spent about 20 h daily in breeding colonies and about

80% of faeces are deposited there (Klimaszyk and Rzymisky 2016). Deposited faeces overload the ecosystem with *N* and *P* by 10^4 to 10^5 times (Garcia et al. 2011). Nutrients transported by breeding birds from water to land ecosystems transform the composition of vegetation, thereby changing hiding and feeding conditions of mammals. Changed chemical properties of soil affect plants (Anderson and Polis 1999), lichens (Zolkos et al. 2013), myxomycetes (Adamonytė et al. 2012), fungal communities (Kutorga et al. 2013), herpetofauna (Kohn and Ota 1991), insects and lizards (Polis and Hurd 1996). Previous studies disclosed how cormorant colonies are reducing relative abundance, diversity and body condition of small mammals, as well as changing their population structure (Balčiauskienė et al. 2014; Balčiauskas et al. 2015). Influence of a great cormorant colony on skull morphometry (Balčiauskienė et al. 2015), stable isotope signatures (Balčiauskas et al. 2016, 2018) and accumulation of chemical elements (Jasiulionis et al. 2018) in small mammals also were observed.

Camera trap based methods are financially cost-effective, they save time resources and are non-invasive (Wearn and Glover-Kapfer 2019). Data collected by camera traps are comparable between different places and in time (Bondi et al. 2010; Silveira et al. 2003). Furthermore, this method enables to record many individual animals. For comparison, tag-based methods can only cover by a small percentage of animals in territory (Rowcliffe et al. 2014). None of the other terrestrial mammal investigation methods (track counts, fecal counts, questionnaire surveys, hunting statistics) could be applied in every ecosystem and for all species bigger than 1 kg weight (Akbaba and Ayas 2012). Camera traps have become a powerful and widely used tool in ecology and conservation. Gysel and Davis began exploring of camera traps as scientific tools in 1956 (Wearn and Glover-Kapfer 2017). In last decades the number of publications is rapidly increasing, from a few per year in 1995–2000 (Rovero et al. 2010) to almost 300 in 2016 (Wearn and Glover-Kapfer 2017). First of all, the camera traps have been used to big animals like bear (Zarzo-Arias 2019), big cat (Cheyne et al. 2013; Garrote et al. 2010;

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Jackson et al. 2006; Karanth et al. 2006) or ungulate (Rovero and Marshall 2009) monitoring. The tool was also used in the studies of medium size mammals like red fox (Diaz-Ruiz et al. 2015; Sarmiento et al. 2009), pine marten (Manzo et al. 2012) and even for the small mammals (Bondi et al. 2010). Camera trap data were used to estimate species richness and site occupancy (Bowler et al. 2017; Hedwig et al. 2017; Rovero et al. 2010; Tobler et al. 2008), in the population ecology (Cheyne et al. 2013), for density estimation using capture-recapture sampling (Garrote et al. 2010; Karanth et al. 2006), and for analysis of the seasonal and daily activity (Caravaggi et al. 2018; Diaz-Ruiz et al. 2015; Ogurtsov et al. 2018; Ribout and Linkie 2009; Rowcliffe et al. 2014).

Seasonal and daily activity of free-living mammals is poorly known as it difficult to quantify activity in the field. There are few ways to study activity of mammals: direct visual observation, radio-collar or GPS telemetry (Ogurtsov et al. 2018). Different methods of testing animal activity give similar results to the camera method (Bondi et al. 2010; Bowler et al. 2017; Lashley et al. 2018; Rovero and Marshall 2009; Rowcliffe et al. 2014).

Therefore, we used camera traps as a tool to monitor mammals in the breeding colonies of great cormorants. The aim of our study was to find what species of mammals visit colonies and to analyse temporal patterns of their visits. We tested working hypothesis that cormorants are affecting seasonal and daily activity of all mammals. Such studies were not done before.

2 Material and methods

2.1 Study sites

2.1.1 The great cormorant colony in Juodkrantė: Great cormorant colony in Juodkrantė (further JCC) is located on the Curonian Spit, Kuršių Nerija National Park (WGS 55° 31' 13.82", 21° 6' 32.57", altitude 5–35 m) (Figure 1). After 100 years of eradication, great cormorants (*Phalacrocorax carbo sinensis*) in JCC started to breed again in 1989. In 2015 the number of breeding pairs in this largest colony in Lithuania was estimated at about 3800 breeding pairs of great cormorants and 340 breeding pairs of grey herons (*Ardea cinerea*) (Knyva, unpublished). The colony is established in the mature pine forest. Most of the cormorant nests are twisted in pine, occasionally in spruce and oak. The area of the colony covers around 14 ha. The active part of the colony, with the highest concentration of nests and the strongest influence of birds, is characterized by scarce herb layer and dying or dead trees. Nitrophilic species of herbs dominate the herb layer, the projection of which is less than 10%. Oldest part of colony is already abandoned by cormorants. Trees in it are dead, many of them rotten, fallen and decaying. Trees and shrubs started re-growing, the herb

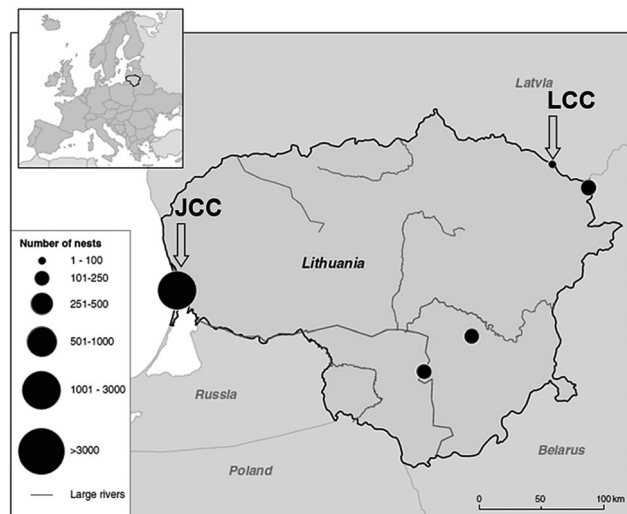


Figure 1: Distribution and size of breeding colonies of the great cormorants (black circles) in Lithuania. Investigated sites marked by arrows: JCC, the great cormorant colony in Juodkrantė; LCC, the great cormorant colony in Lukštas lake.

layer is re-establishing. Among re-growth deciduous trees (*Quercus robur*, *Betula pubescens*, *Alnus glutinosa*) dominate (Matulevičiūtė et al. 2018).

2.1.2 The great cormorant colony in Lukštas: Great cormorant colony in Lukštas Lake (further LCC) is located in East Lithuania, Zarasai district (WGS 55° 51' 0.41", 26° 12' 4.67", altitude 140 m) and situated in peninsula at the northern part of lake (Figure 1). 130 breeding pairs were counted in 2016. Most nests are twisted in black Alder, pine and birch trees. Grey herons are not present in the colony. The area of the colony covers around 1 ha.

2.2 Camera trapping

The activities of mammals were evaluated using camera traps (system with a motion trigger/sensor that activates a camera to take a photograph when an animal is present). We deployed camera traps in three points (one camera per point) in the JCC and one point in LCC.

Camera trapping in JCC lasted from November 2014 to September 2015 and from June 2017 to October 2018 involving 1376 trap days of effort. Camera trapping in LCC lasted from February 2015 to February 2016 and from November 2016 to July 2019 involving 1317 trap days of effort (Supplementary Table S1).

In this study we used Ltl Acorn-5210A cameras. Sensors were set to the maximum sensitivity. The trigger interval was set at 0 s. We used trail-targeted (passive camera) trapping design with no attractants. Cameras were mounted on the trees 40–60 cm above ground and checked (replacing batteries and SD memory cards) 3–4 times during the year. Time and date were automatically recorded on each picture. Pictures taken with at least 1-hour interval were treated as independent observations (trap events). Camera day lasted from 00:00 to 23:59. Year was divided into periods of the cormorant breeding time (BT) and non-breeding time (NT). BT lasted from April (when chicks

begin to hatch from eggs) to July (when chicks finally leave the cormorant area). Non-breeding time was the period from August to March.

Each day was divided into three periods: daytime, nighttime and twilight. Twilight (T) was defined as a 60 min period before and after sunrise/sunset, so that each day had four twilight hours. Daytime (D) was a period between 60 min after sunrise and 60 min before sunset. Night (N) was a period between 60 min after sunset and 60 min before sunrise. Time of sunset and sunrise were estimated using the website www.day.lt.

Some devices failed to record time; for this reason, we lost information about day activity in 117 trap events from JCC.

2.3 Diversity assessment and statistical analysis

In this study we used data on relative animal registrations with respect to the fact that individuals may be registered several times in the same colony (Anderson et al. 2000). Relative shooting frequency (RSF, photos/100 days) was calculated by formula: $RSF = (TE/TD) * 100$, where TE (trap events) = number of independent photographs; TD (effort of trapping days) = number of camera trapping days. For RSF values 95% confidence intervals (further – CI) were used. RSF and 95% CI for RSF were calculated in OpenEpi software (Dean et al. 2013), using Wilson method of the score interval (Brown et al. 2001). If the lower range of CI was negative, RSF was considered non-significant. Following Nakagawa and Cuthill (2007) and Wasserstein and Lazar (2016), comparing RSF differences effect size was calculated; it was expressed according adjusted Cohen's w (Thomas et al. 1991). Mammal diversity in the cormorant colonies was evaluated using Shannon's diversity index H ($\log_2$ base), according to (Krebs 1999).

3 Results

3.1 Diversity of mammals in the great cormorant colonies

During 1317 trap days 668 independent photographs (trap events) were taken and 9 species of mammals were

detected in LCC: red fox (*Vulpes vulpes*), raccoon dog (*Nyctereutes procyonoides*), roe deer (*Capreolus capreolus*), moose (*Alces alces*), wild boar (*Sus scrofa*), hare (*Lepus* sp.), red squirrel (*Sciurus vulgaris*), marten (*Martes* sp.) and European badger (*Meles meles*). In JCC 895 independent photographs were taken during 1376 trap days and 9 species of mammals were registered: *S. scrofa*, *C. capreolus*, *A. alces*, *V. vulpes*, *N. procyonoides*, *S. vulgaris*, *Lepus* sp., raccoon (*Procyon lotor*), and sika deer (*Cervus nippon*) (Table 1, Figure 2).

3.2 Seasonal activity of mammals in cormorant colonies

Moose in LCC was registered only once in February (Supplementary Table S2). JCC in wintertime was not attractive for moose, however, from April onwards, moose registration frequencies increased with the peak of activity in August (RSF = 28.33; 95% CI = 21.04–36.97, Supplementary Table S3).

Roe deer registration frequencies did not differ between seasons in both cormorant colonies. In LCC roe deer RSF was 6.22 (95% CI = 4.37–8.79) in BT and 8.7 (6.98–10.78) photos/100 days in NT (Cohen's $w = 0.039$, no effect) while in JCC respective values were 25.67 (21.99–27.93) and 21.82 (19.32–24.66) photos/100 days (Cohen's $w = 0.031$, no effect).

Wild boars visited LCC mainly in July, RSF = 12.75 (95% CI = 7.60–20.59) and December, 12.9 (8.10–19.94), average RSF = 2.66 (1.86–3.79) photos/100 days in other months (effect size small, $w = 0.244$). RSF of wild boar in BT was 6.65 (95% CI = 4.73–9.29) and differed from that in NT, RSF = 3.17 (CI 2.92–8.99) photos/100 days (Supplementary Table S2), Cohen's $w = 0.107$, effect size small.

Table 1: Relative shooting frequency (RSF, photos/100 days) and diversity indexes in the great cormorant colonies in Lukštas Lake (LCC) and Juodkrantė (JCC).

Species	LCC, RSF (CI 95%)	JCC, RSF (CI 95%)	Cohen's w	Effect size
<i>Alces alces</i>	0.08 (0.01–0.43)	8.15 (6.81–9.70)	0.272	Small
<i>Capreolus capreolus</i>	7.82 (6.49–9.40)	23.18 (21.03–25.49)	0.241	Small
<i>Cervus nippon</i>	–	1.16 (0.77–1.88)	0.106	Small
<i>Sus scrofa</i>	4.4 (3.42–5.65)	25.94 (23.70–28.33)	0.348	Medium
<i>Vulpes vulpes</i>	18.6 (16.59–20.79)	4.94 (3.92–6.22)	0.298	Small
<i>Nyctereutes procyonoides</i>	18.53 (16.52–20.72)	0.29 (0.11–0.75)	0.405	Medium
<i>Procyon lotor</i>	–	0.51 (0.25–1.05)	0.070	No effect
<i>Lepus</i> sp.	0.84 (0.47–1.49)	0.15 (0.04–0.53)	0.070	No effect
<i>Sciurus vulgaris</i>	0.15 (0.04–0.55)	0.73 (0.40–1.33)	0.043	No effect
<i>Martes</i> sp.	0.23 (0.08–0.67)	–	0.048	No effect
<i>Meles meles</i>	0.08 (0.01–0.43)	–	0.028	No effect
Shannon's H	1.99	2.00		

Between-site differences expressed as effect size based on Cohen's w .

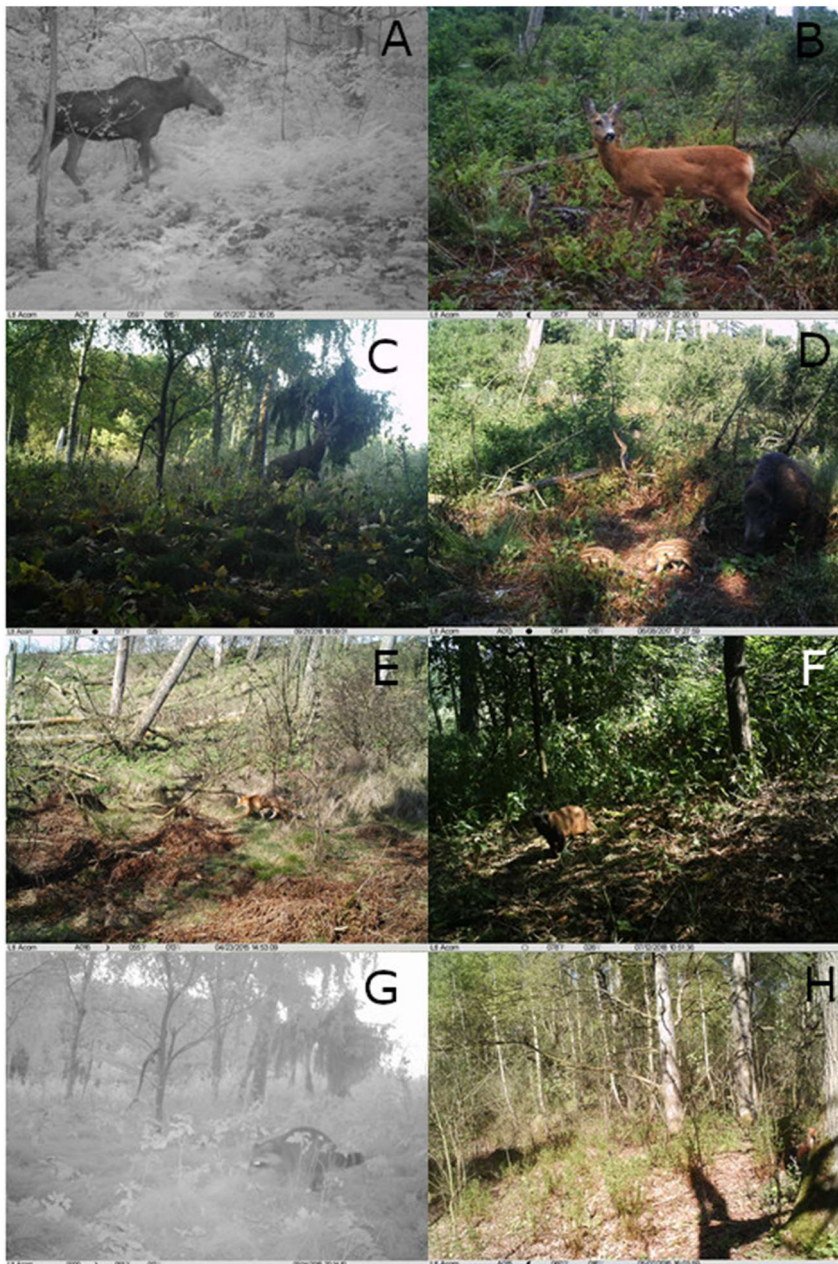


Figure 2: Mammals registered by camera traps in great cormorants colonies (A) *Alces alces*; (B) *Capreolus capreolus*; (C) *Cervus nippon*; (D) *Sus scrofa*; (E) *Vulpes vulpes*; (F) *Nyctereutes procyonoides*; (G) *Procyon lotor*; (H) *Sciurus vulgaris*.

In JCC wild boars were most active in June, August and September: 21.01 (CI = 15.05–28.55), 87.5 (80.40–92.28) and 153.54 (145.00–159.20) photos/100 days, respectively, while average RSF in other months 6.97 (5.56–8.70) photos/100 days was significantly lower ($w = 0.720$, large effect size) (Supplementary Table S3; Figure 3D). In JCC wild boar relative shooting frequency in BT, RSF = 9.86 (CI 7.51–12.87), was lower than in NT, RSF = 34.76 (31.70–37.95) photos/100 days (confidence intervals not overlapping, Cohen's $w = 0.277$, small effect size).

Red fox registrations in LCC were high from April, with RSF = 40.0, 95% CI = 31.68–48.94 to August, RSF = 23.66,

CI 16.17–33.23, with peak of activity in July RSF = 75.49, CI = 66.32–82.81 photos/100 days (Supplementary Table S2; Figure 3A). Red fox was visiting LCC much more frequently in BT (RSF = 41.85.1, CI = 37.45–46.37) than in NT (RSF = 5.88, CI 4.48–7.66 photos/100 days, $w = 0.487$, medium effect size). Highest red fox RSF in JCC was in June, 13.77 (8.99–20.50) photos/100 days, compared to 3.96 (3.01–5.19) photos/100 days in other months ($w = 0.174$, small effect size; Supplementary Table S3; Figure 3C). Differences due to breeding were significant, RSF = 7.80 (CI 5.74–10.53) photos/100 days in BT, compared to RSF = 3.37 (CI 2.37–4.78) in NT, though effect size is small (Cohen's $w = 0.090$).

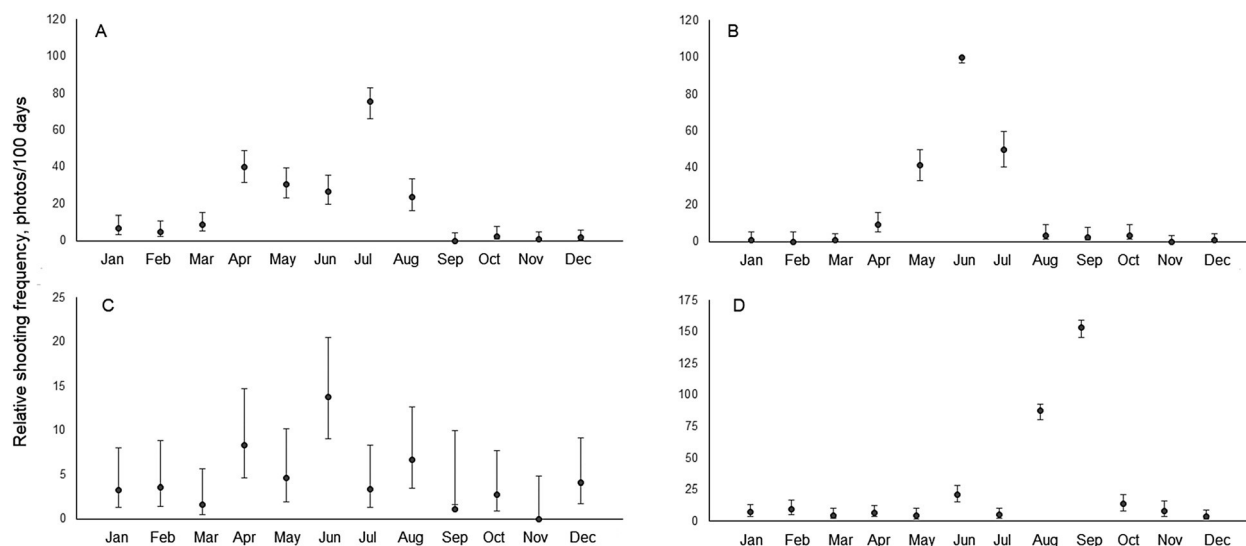


Figure 3: Seasonal activity of *Vulpes vulpes* (A) and *Nyctereutes procyonoides* (B) in the colony of great cormorants in Lukštas Lake (LCC), *V. vulpes* (C) and *Sus scrofa* (D) in the colony of great cormorants in Juodkrantė (JCC). 95% confidence interval is denoted by whiskers.

In LCC raccoon dog RSF was highest in breeding time, April (RSF = 9.17, CI = 5.20–15.67) to July (50.0 and 40.47–59.53, respectively) with peak of activity in June (RSF = 100.0, CI = 96.90–100.0 photos/100 days) (Supplementary Table S2; Figure 3B). In LCC, raccoon dog RSF was 50.0 (CI = 45.48–54.52) photos/100 days in BT, but just 1.29 (CI = 0.54–2.59) photos/100 days in NT. This difference is significant with large effect size ($w = 0.691$). In JCC raccoon dogs were photographed only 4 times (Supplementary Table S3).

3.3 Daily activity patterns of mammals

Without respect of the cormorant presence, moose (JCC 47.2% of all TE, CI = 38.1–56.6%), roe deer (LCC 50.5%, CI = 41.0–60.0%; JCC 53.2%, CI = 46.8–59.5%) and red fox

(LCC 57.1%, CI = 50.9–63.2%; JCC 54.4%, CI = 41.6–66.6%) were mostly active by day. Wild boar in LCC was mostly active by night (58.6%, CI = 45.8–70.4%) and in JCC by day (43.1%, CI = 38.0–48.4%). Raccoon dog in LCC was mostly active by day (56.6%, CI = 50.3–62.6%). Distribution of the hourly activities of mammals in LCC and JCC are given in Tables S4 and S5.

Breeding of the cormorants, however, brought changes in the daily activity patterns of most of the species (Table 2). In LCC wild boar during NT was active only at night (100% of TE). However, during the BT highest frequency of registrations of the wild boar was in the daytime (61.3%, CI = 48.2–76.3%) of all TE ($w = 0.325$, medium effect size).

In JCC activity peak of wild boar during NT was at daytime (46.1% of all TE, CI = 40.5–51.8%). During BT activity switched to twilight (39.1%, CI = 26.4–53.5%) and

Table 2: Frequency of trap events (%) of *Sus scrofa*, *Vulpes vulpes* and *Nyctereutes procyonoides* at the great cormorant colonies in Lukštas Lake (LCC) and Juodkrantė (JCC) throughout periods of the day in cormorant breeding time (BT) and non-breeding time (NT).

Season	Time	LCC			JCC		
		<i>S. scrofa</i>	<i>V. vulpes</i>	<i>N. procyonoides</i>	<i>S. scrofa</i>	<i>V. vulpes</i>	<i>N. procyonoides</i>
BT	N	22.6 (11.4–39.8)	22.8 (17.6–29.0)	19.2 (14.7–24.8)	37.0 (24.5–51.4)	16.7 (7.9–31.9)	100
	T	16.1 (7.1–32.6)	18.0 (13.3–23.8)	22.7 (17.8–28.4)	39.1 (26.4–53.5)	33.3 (20.2–49.7)	–
	D	61.3 (43.8–76.2)	59.2 (52.4–65.7)	58.1 (51.7–64.3)	23.9 (13.9–37.9)	50.0 (34.5–65.5)	–
NT	N	100	30.8 (18.6–46.2)	70.0 (39.9–89.2)	25.4 (20.8–30.7)	28.6 (13.8–50.0)	100
	T	0	23.1 (12.7–38.4)	10.0 (1.8–40.4)	28.5 (23.6–33.9)	9.5 (2.7–28.9)	–
	D	0	46.2 (31.6–61.4)	20.0 (5.7–51.0)	46.1 (40.5–51.8)	61.9 (40.9–79.3)	–

D, daytime; N, night; T, twilight. CI shown in parentheses.

nighttime (37.0%, CI = 24.5–51.4%), however, these changes were not significant (Cohen's w 0.035 and 0.047, respectively, no effect).

In LCC daytime red fox activity was higher during BT (59.2% of all TE, CI = 52.4–65.7%) than in NT (46.2%, CI = 31.6–61.4%), however, not significantly (w = 0.059, no effect). In JCC, on the contrary, daytime red fox activity was higher during NT (61.9% of all TE, CI = 40.9–79.3%) than in BT (50.0%, CI = 34.5–65.5%, Table 2). These activity changes also had no significant effect (w = 0.053).

Raccoon dog daily activity peaks in LCC were also related to cormorant breeding time (Table 2). In BT raccoon dog activity at nighttime (19.2% of all TE, CI = 14.7–24.8%) was lower than at nighttime in NT (70.0% of all TE, CI = 39.7–89.2%, w = 0.027, no effect). In JCC, raccoon dog was registered in the nighttime only (Table 2).

4 Discussion

Following global development of the population of the great cormorants in the last decades, ecological impacts were investigated by many authors (Goc et al. 2005; Gwiazda et al. 2010; Hobara et al. 2005; Klimaszyk and Rzymisky 2016; Klimaszyk et al. 2015). Cormorants not only overload territory with nutrients and transform the environment. Part of fish brought to the colony as food for chicks is being lost (Pūtys 2012). Furthermore, during the cormorants breeding season, some chicks fall out of their nests. Lost fish and dropped chicks becoming easy prey for scavenging mammals. Changed environment affects nearly all aspects of the small mammals (Balčiauskienė et al. 2014; Balčiauskas et al. 2015), however changes in mammalian activity were not investigated. We present here first results of investigation of activity patterns of mammal in the colonies of great cormorants.

In the cormorant breeding season the intensity of the predator visits was greatly altered. We found, that in BT red fox visit frequency increased 2–7 times, compared to NT. In July, fox RSF was 75.5 photos/100 days, which is 12.5 times more frequent than in NT (see Supplementary Table S2). In LCC, raccoon dog activity in BT increased 38.5 times (see Supplementary Table S2, Figure 3B).

Activity of the omnivorous wild boar was also affected by cormorants in JCC, visits to the area increased almost 3 times in BT (June), being, however, less intensive as in August and September, when wild boars were coming for acorns (see Supplementary Table S3; Figure 3D). In LCC

with no oak trees, wild boar visits were over two times more frequent in BT (see Supplementary Table S2).

Daily activity patterns in mammals are regulated by external factors such as food accessibility (Diaz-Ruize et al. 2015; Masi et al. 2009). Cormorants feed their chicks in the daylight (Coleman and Richmond 2007), so it is time when they lose some of the prey. We found, that predators (foxes and raccoon dogs) and omnivores (wild boars) adapted to the cormorant activity during the day.

Other authors (Lee et al. 2019; Marques and Fabian 2019; Ogurtsov et al. 2018) pointed out that most mammals are mostly active at twilight and nighttime. Carnivore mammals are mainly characterised as nocturnal, as the nighttime is more successful time to find prey and avoid contact with humans (Georgiev et al. 2015; Lee et al. 2019). Raccoon dog also exhibits nocturnal activity (Akbaba and Ayas 2012; Ikeda et al. 2016; Zoller and Drygala 2013). Ogurtsov et al. (2018) reported 64% trap events of raccoon dog at nighttime. Raccoon dog activity in NT was in agreement with these findings, as 85.7% of all TE in cormorant colonies were recorded in nighttime. However, availability of accessible food switched raccoon dogs to daytime activity: during BT 58.1% of registrations were in the daytime.

In England, Northern Ireland and Spain red foxes are categorized as a nocturnal (Caravaggi et al. 2018; Diaz-Ruiz et al. 2015; Doncaster and Macdonald 1997). In Turkey red fox was randomly active during the day and frequently active from 20:00 to 02:00 (Akbaba and Ayas 2012). Also, this predator categorized as a nocturnal in Bulgaria with a small activity peak in the daytime from 7:00 to 9:00 and from 12:00 to 14:00 (Georgiev et al. 2015). Servin et al. (1991) suggested that fox females are more active during the night than during the day then males are active during both day and night. However red fox has great behavioral plasticity in activity patterns which allows them to adapt to prey availability or other environmental changes (Diaz Ruize et al. 2015). In Japan red fox was categorized as a cathemeral (active throughout the day) animal (Ikeda et al. 2016).

Findings of our study support this view as in both investigated cormorant colonies fox was active in daytime (see Table 2). Differences were related to cormorant breeding season: in BT fox was more active at daytime in LCC (see Table 2). On the contrary, in JCC, fox was less active in daytime in BT compared to NT. We see here human-induced disturbance as a factor, influencing fox behavior. Curonian Spit, where JCC is situated, is one of the

most visited territories in Lithuania. The cormorant colony is a popular tourist destination and has a viewing tower. At day, nearly all time tourists are present and disturbing foxes. On the contrary, LCC is difficult to access, so people do not visit it. Changes in animal activity patterns under human disturbance are confirmed by other authors (Doormal et al. 2015; Lee et al. 2019).

Wild boar activity is characterised as nocturnal (Caruso et al. 2018; Ogurtsov et al. 2018). In the cormorant colonies changes of activity were observed: in LCC wild boar was more active at daytime (61.3% of all TE) in BT, while in NT at nighttime (see Tables 2 and S4). In JCC activity patterns of wild boar were additionally influenced by human disturbance. In BT, when many visitors are registered, wild boar was active at nighttime and at twilight. In NT, when the number of visitors is minimal, wild boar was active at daytime (see Tables 2 and S5).

Thus, colonies of great cormorants were influencing seasonal and daily activity of mammals, mostly predators (raccoon dog and red fox) and omnivorous wild boar. These behavioral changes make it possible to replenish foods with lost fish and nesting cormorant juveniles. Cormorant activity did not affect roe deer activity in both investigated colonies. In conclusion:

- Breeding activity of cormorants is most likely to alter the daily activity of raccoon dog, red fox and wild boar with various degree of significance.
- In LCC average relative shooting frequency of these carnivores in the breeding season was a few times higher than in non-breeding time.
- In JCC average relative shooting frequency of red fox in BT was 2 times higher than in NT.

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Conflict of interest statement: The authors declare no conflicts of interest regarding the study and article.

Research ethics: The study was conducted in accordance with Lithuanian (the Republic of Lithuania Law on the Welfare and Protection of Animals No. XI-2271) and European legislation (Directive 2010/63/EU) on the protection of animals. According to the Minister of the Environment and the Director of the State Food and Veterinary Service, the Order No. D1-533/B1-310 requirements, wild animals living in the wild may be photographed, filmed or observed (without disturbing them) without the permit from the Environmental Protection Agency. This applies to non-protected species.

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